

Ex. No. 9	Use Process Explorer to identify suspicious processes
Date	

Aim:-

To use Process Explorer to identify suspicious processes on a Windows system by checking their signatures, paths, resource usage, and network activity.

Description:

Process Explorer is a powerful Windows tool that provides detailed information about running processes, allowing users to monitor, troubleshoot, and detect suspicious activities. It can be used to detect potential malware or harmful processes by analyzing various aspects of running applications.

Here's a step-by-step guide on how to use Process Explorer to identify suspicious processes on your system:

Project Steps:

Step 1: Download and Set Up Process Explorer

1. Download Process Explorer:

- Go to the official Microsoft Sysinternals website and download Process Explorer: [Download Process Explorer](#).

2. Extract the Program: ○ Once downloaded, extract the ZIP file to a folder.

3. Run Process Explorer:

- Open the folder where you extracted the files and run procexp64.exe (64-bit) or procexp.exe (32-bit) as Administrator by right-clicking and selecting "Run as Administrator."

Step 2: Familiarize Yourself with the Interface

When Process Explorer starts, you'll see an interface that lists all running processes, showing their hierarchy, memory usage, CPU usage, and much more.

1. Process Tree:

- The main window shows a tree of all running processes, with each child process listed under its parent.
- Processes are color-coded to indicate their status:

- **Pink:** Suspended processes.
- **Light Blue:** Processes running under the same user as you.
- **Dark Blue:** Services or processes running under system accounts.
- **Green:** New processes (appearing and disappearing quickly may be suspicious).
- **Red:** Processes that have just exited.

2. Columns:

- Various columns show information such as Process ID (PID), CPU usage, Memory usage, and more.

Step 3: Identify Suspicious Processes

Now that you are familiar with the interface, follow these steps to investigate potentially suspicious processes:

1. Look for Unfamiliar Processes:

- Scan the list of running processes and identify any names that you don't recognize. Some malware disguises itself with legitimate-looking names, but odd names can be a red flag.
- Legitimate processes should typically be from trusted sources (e.g., Microsoft, Intel, Adobe, etc.).

2. Verify Digital Signatures:

- Right-click any suspicious process and select Properties.
- Go to the Image tab and check if the process has a valid Digital Signature.
- Legitimate applications from trusted developers will have valid signatures. Processes without signatures or with invalid signatures are suspicious.

3. Check the Path of the Process:

- In the Properties window of the suspicious process, look under the Image tab for the Path.
- Verify if the process is located in a legitimate directory. For example, legitimate system processes should reside in C:\Windows\System32.
- If the process is running from a temporary folder, user folder, or a random directory, it might be malware.

4. Monitor CPU, Memory, and Disk Usage:

- Look at the CPU, Memory, and Disk usage columns.

- A process that is using an unusual amount of resources without explanation could be malicious. For example, high CPU usage on a process like svchost.exe can indicate malware.

5. Check Process Description and Company Name:

- Look at the Description and Company Name columns in Process Explorer.
- Legitimate processes usually provide clear descriptions and are tied to well-known companies.
- Lack of information or suspicious company names should prompt further investigation.

6. Check the Process Network Activity:

- Some malware communicates with external servers. You can inspect a process's network connections:
 - Right-click on the process and select Properties, then go to the TCP/IP tab to see network activity associated with the process.
 - Unexpected or unexplained external connections can indicate a malicious process.

Step 4: Perform a Google Search on Suspicious Processes

If you find a process that looks suspicious or unfamiliar:

1. Search online: Perform a quick Google search using the process name (e.g., processname.exe) to find out whether the process is known to be safe or malicious.
2. Check Malware Databases: Use online databases such as VirusTotal or ProcessLibrary to verify whether the process is known for being part of malware.

Step 5: Kill or Suspend Suspicious Processes If

you find a process that you believe is malicious:

1. Kill the Process:

- Right-click the suspicious process and select Kill Process to terminate it immediately.

2. Suspend the Process:

- If you are unsure and want to prevent the process from causing damage while you investigate further, right-click and select Suspend.

3. Remove the Source File:

- Go to the location of the file (using the Path from the Properties window) and delete the executable file if confirmed to be malware.

- Be cautious, as some malware might prevent you from deleting it. In such cases, you may need to boot into Safe Mode or use a specialized tool.

Step 6: Scan Your System

After terminating any suspicious processes, it's a good idea to:

- Run a full antivirus scan using trusted antivirus software.
- Use Malware Removal Tools like Malwarebytes or Windows Defender to perform an in-depth scan of your system.

Example of Identifying a Suspicious Process

1. You notice a process called randomname123.exe using an unusually high amount of CPU.
2. Right-click it, select Properties, and find it is running from a suspicious folder, not the system folder.
3. You check the TCP/IP tab and see it is making external connections to unknown IP addresses.
4. A quick search shows that this process is often associated with a specific type of malware.
5. You kill the process and remove the executable file from the system.

Output Screenshots:

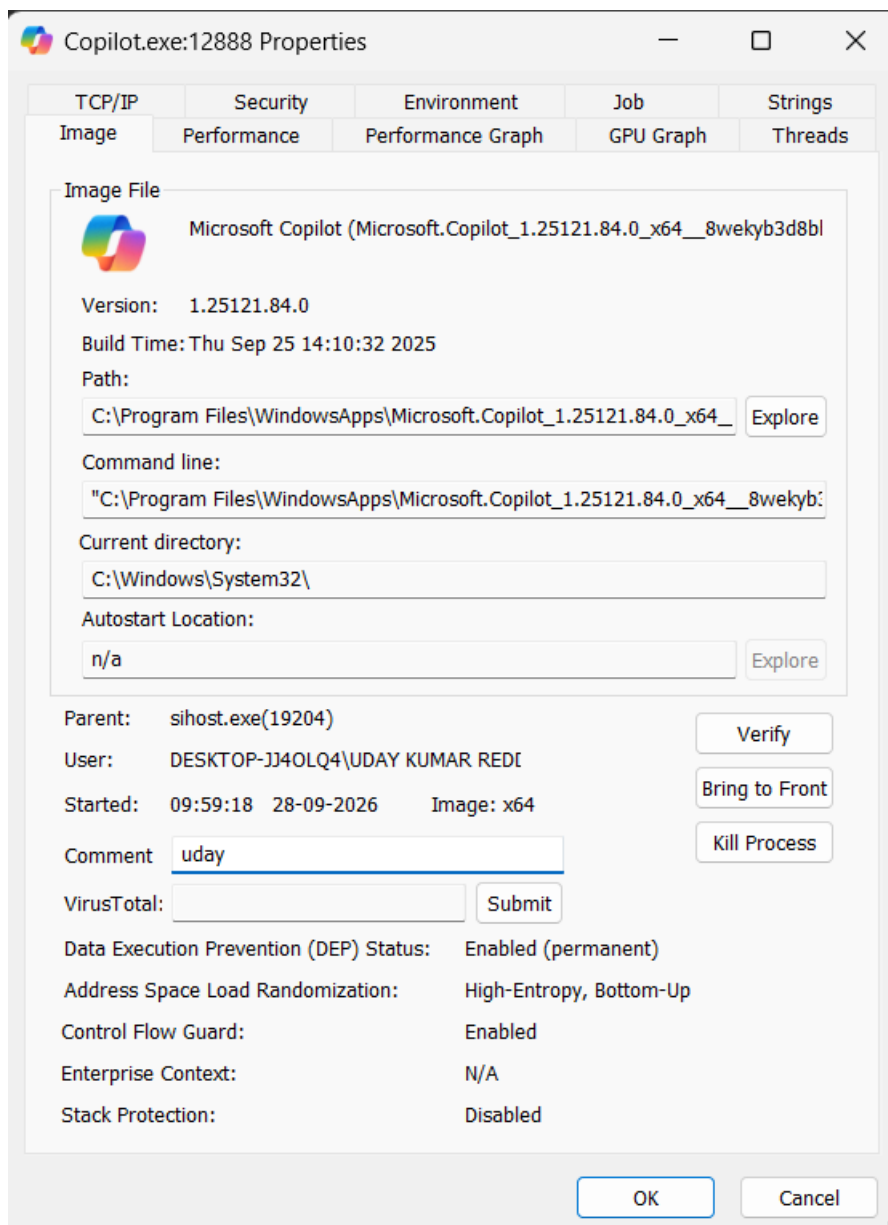
Process Explorer - Sysinternals: www.sysinternals.com [DESKTOP-JJ4OLQ4\UDAY KUMAR REDDY]

File Options View Process Find Users Help

<Filter by name>

Process	CPU	Private Bytes	Working Set	PID	Description	Company Name
Secure System		176 K	62,816 K	236		
Registry		6,132 K	45,032 K	280		
System Idle Process	92.69	60 K	8 K	0		
System	0.29	44 K	1,344 K	4		
Interrupts	0.29	0 K	0 K	n/a	Hardware Interrupts and DPCs	
smss.exe		1,172 K	1,596 K	924		
Memory Compression	< 0.01	976 K	2,00,156 K	4036		
csrss.exe	< 0.01	2,788 K	7,544 K	1336		
wininit.exe		1,836 K	9,788 K	1428		
services.exe	0.19	6,208 K	12,644 K	1576		
svchost.exe	< 0.01	13,596 K	43,772 K	1736	Host Process for Windows S...	Microsoft Corporation
unsecapp.exe		1,972 K	9,536 K	7712		
WmiPrvSE.exe		24,428 K	33,080 K	7960		
SearchHost.exe	< 0.01	69,556 K	1,68,500 K	9704		Microsoft Corporation
msedgewebview2.exe		42,776 K	1,33,976 K	18816	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		2,984 K	13,436 K	17744	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe	< 0.01	63,784 K	83,916 K	14484	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		15,800 K	46,012 K	10816	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		8,636 K	20,712 K	19264	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		68,232 K	1,17,772 K	10256	Microsoft Edge WebView2	Microsoft Corporation
StartMenuExperienceHost.exe	< 0.01	1,05,980 K	1,90,008 K	14264	Windows Start Experience H...	Microsoft Corporation
WidgetBoard.exe		5,312 K	33,656 K	7400		Microsoft Corporation
RuntimeBroker.exe		9,340 K	56,184 K	9400	Runtime Broker	Microsoft Corporation
WidgetService.exe		5,804 K	29,284 K	20360	WidgetService.exe	Microsoft Corporation
MicrosoftStartFeedProvi...		10,316 K	43,476 K	12224		Microsoft Corporation
TextInputHost.exe		1,16,524 K	2,05,836 K	10736		Microsoft Corporation
RuntimeBroker.exe		3,164 K	22,816 K	10496	Runtime Broker	Microsoft Corporation
AppActions.exe		12,660 K	43,360 K	16600	AppActions	Microsoft Corporation
ApplicationFrameHost.e...		55,136 K	67,304 K	17680	Application Frame Host	Microsoft Corporation
SystemSettings.exe	Susp...	1,01,448 K	4,144 K	18148	Settings	Microsoft Corporation

CPU Usage: 6.37% Commit Charge: 50.31% Processes: 227 Physical Usage: 49.94%



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Process	CPU	Private Bytes	Working Set	PID	Description	Company Name
svchost.exe		1,920 K	9,508 K	2472	Host Process for Windows S...	Microsoft Corporation
svchost.exe		6,964 K	17,644 K	2480	Host Process for Windows S...	Microsoft Corporation
taskhost.exe	< 0.01	8,876 K	25,204 K	6028	Host Process for Windows T...	Microsoft Corporation
RtkAudUServiceF4.exe	< 0.01	26,764 K	7,628 K	16028	Realtek HD Audio Universal...	Realtek Semiconductor
IntelCpHDCPSvc.exe		1,476 K	7,660 K	2504	Intel HD Graphics Drivers fo...	Intel Corporation
svchost.exe	< 0.01	3,900 K	12,608 K	2516	Host Process for Windows S...	Microsoft Corporation
svchost.exe		3,104 K	12,000 K	2524	Host Process for Windows S...	Microsoft Corporation
svchost.exe		1,640 K	7,424 K	2612	Host Process for Windows S...	Microsoft Corporation
WUDFHst.exe		1,960 K	9,040 K	2654		
svchost.exe	< 0.01	12,684 K	38,312 K	2708	Host Process for Windows S...	Microsoft Corporation
svchost.exe		2,616 K	14,752 K	2724	Host Process for Windows S...	Microsoft Corporation
svchost.exe	< 0.01	2,324 K	10,176 K	3008	Host Process for Windows S...	Microsoft Corporation
svchost.exe	< 0.01	3,456 K	16,420 K	3060	Host Process for Windows S...	Microsoft Corporation
sihost.exe	< 0.01	8,224 K	51,168 K	19204	Shell Infrastructure Host	Microsoft Corporation
CrossDeviceResume...		35,156 K	64,680 K	17144		Microsoft Corporation
Copilot.exe		73,344 K	2,17,480 K	12888	Microsoft Copilot	Microsoft Corporation
WhatsApp.Root	< 0.01	1,35,752 K	3,32,592 K	2696	WhatsApp Root	
mseedgewebview2...		1,01,328 K	1,42,584 K	20880	Microsoft Edge WebView2	Microsoft Corporation
mseedgewebvie...		2,940 K	13,856 K	492	Microsoft Edge WebView2	Microsoft Corporation
mseedgewebvie...		2,25,508 K	1,53,468 K	3428	Microsoft Edge WebView2	Microsoft Corporation
mseedgewebvie...		15,360 K	28,752 K	14884	Microsoft Edge WebView2	Microsoft Corporation
mseedgewebvie...		8,924 K	8,364 K	5824	Microsoft Edge WebView2	Microsoft Corporation
mseedgewebvie...	< 0.01	5,35,664 K	5,20,256 K	3940	Microsoft Edge WebView2	Microsoft Corporation
mseedgewebvie...		9,008 K	11,892 K	9132	Microsoft Edge WebView2	Microsoft Corporation
ShellHost.exe	< 0.01	33,604 K	67,296 K	8112	ShellHost	Microsoft Corporation
svchost.exe	< 0.01	3,676 K	14,632 K	3172	Host Process for Windows S...	Microsoft Corporation
svchost.exe		1,696 K	9,040 K	3556	Host Process for Windows S...	Microsoft Corporation
SynTPEnhService.exe		3,972 K	13,096 K	3564	64-bit Synaptics Pointing Enh...	Synaptics Incorporated
SynTPEnh.exe		8,440 K	27,228 K	20008	Synaptics TouchPad 64-bit E...	Synaptics Incorporated
svchost.exe	< 0.01	19,388 K	22,980 K	3700	Host Process for Windows S...	Microsoft Corporation
svchost.exe		2,164 K	10,456 K	3764	Host Process for Windows S...	Microsoft Corporation
svchost.exe	< 0.01	1,524 K	7,496 K	3772	Host Process for Windows S...	Microsoft Corporation
svchost.exe		3,508 K	19,328 K	3780	Host Process for Windows S...	Microsoft Corporation
svchost.exe		2,224 K	11,788 K	4076	Host Process for Windows S...	Microsoft Corporation
svchost.exe		2,408 K	11,592 K	3488	Host Process for Windows S...	Microsoft Corporation
NetworkCap.exe		6,872 K	34,232 K	3436		HP Inc.
AppHelperCap.exe		6,588 K	34,780 K	3384		HP Inc.
SysInfoCap.exe		10,620 K	39,320 K	3536		HP Inc.
TouchpointAnalyticsClients...		67,484 K	72,744 K	3528	HP Insights Analytics Service	HP Inc.
svchost.exe		2,048 K	9,920 K	3620	Host Process for Windows S...	Microsoft Corporation
svchost.exe		4,504 K	57,908 K	4296	Host Process for Windows S...	Microsoft Corporation
svchost.exe		20,456 K	27,344 K	20560		
audiofsg.exe			25,016 K	4492	Host Process for Windows S...	Microsoft Corporation
svchost.exe	0.10	13,316 K	22,580 K	5004	Host Process for Windows S...	Microsoft Corporation
svchost.exe		9,004 K				

Copilot.exe:12888 Properties

Image TCP/IP
Performance Security
Performance Graph Environment
GPU Graph Job
Threads Strings

☒ Resolve addresses

Protocol	Local Address	Remote Address	State
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OK Cancel

CPU Usage: 3.34% Commit Charge: 49.00% Processes: 227 Physical Usage: 48.83%

RuntimeBroker.exe:19220 Properties

Image Performance Performance Graph GPU Graph Threads TCP/IP Security Environment Job Strings



User: DESKTOP-JJ4OLQ4\UDAY KUMAR REDDY
 SID: S-1-5-21-1185852721-1740634740-4041162602-1001
 Session: 5 Logon Session: 3a004c8
 Virtualized: No Protected: No

Group	Flags
BUILTIN\Administrators	Deny
BUILTIN\Users	Mandatory
CONSOLE LOGON	Mandatory
DESKTOP-JJ4OLQ4\None	Mandatory
Everyone	Mandatory
LOCAL	Mandatory
Mandatory Label\Medium Mandatory Level	Integrity
NT AUTHORITY\Authenticated Users	Mandatory
NT AUTHORITY\INTERACTIVE	Mandatory
NT AUTHORITY\Local account	Mandatory
NT AUTHORITY\Local account and member of Administrators group	Deny
NT AUTHORITY\LogonSessionId_0_60818182	Mandatory
NT AUTHORITY\NTLM Authentication	Mandatory
NT AUTHORITY\This Organization	Mandatory

Group SID: S-1-2-0



Privilege	Flags
SeChangeNotifyPrivilege	Default Enabled
SeIncreaseWorkingSetPrivilege	Disabled
SeShutdownPrivilege	Disabled
SeTimeZonePrivilege	Disabled
SeUndockPrivilege	Disabled

Rubrics:

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1.GitHub Activity & Submission Regularity	3	
2.Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

Result: Process Explorer was used to inspect running processes and flag suspicious ones, which were then verified with online malware databases and terminated.